

LAB EXPERIMENT RECORD

SET 13: Geographic Data

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Date	10 october 2026
Day	thursday

Aim

To visualize the geographic distribution of cities using population and temperature information.

Dataset

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

PROGRAM 1: Create a map chart

To visualize the geographic distribution of cities using population and temperature information.

```
# # Set 13 Q1 - Geographic Map
city <- c("City A", "City B", "City C")
population <- c(500000, 700000, 600000)
temperature <- c(75, 68, 80)
# Sample coordinates for visualization
longitude <- c(80, 82, 84)
latitude <- c(13, 14, 12)
plot(
  longitude,
  latitude,
  type = "n",
  main = "Geographic Distribution of Cities",
  xlab = "Longitude",
  ylab = "Latitude"
)
points(
  longitude,
```

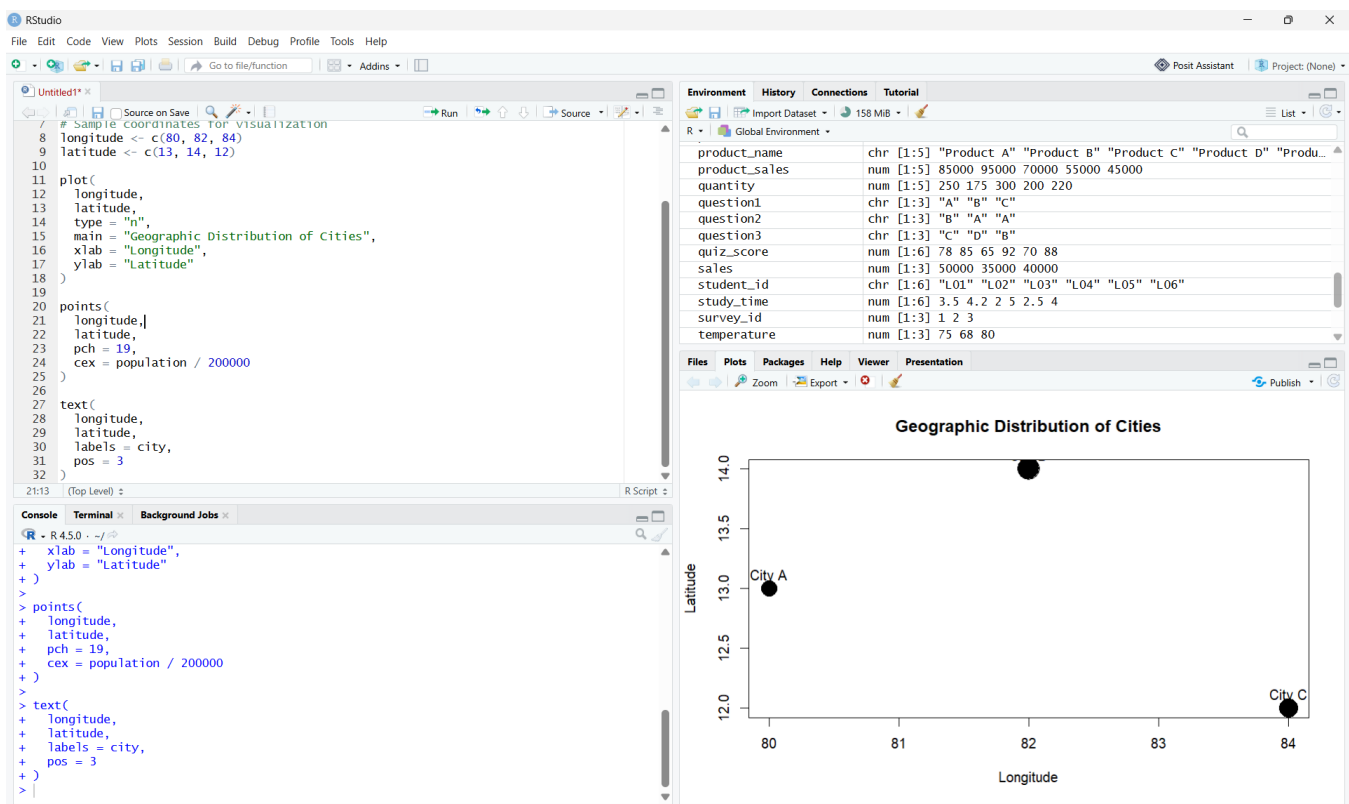
```

latitude,

pch = 19,

cex = population / 200000
)
text(
  longitude,
  latitude,
  labels = city,
  pos = 3
)

```



Output 1: Monthly Sales Line Chart

Observation:

The three cities are displayed according to their sample latitude and longitude positions. The size of each point represents population.

PROGRAM 2: Temperature vs Population Scatter Plot

To study the relationship between average temperature and population size..

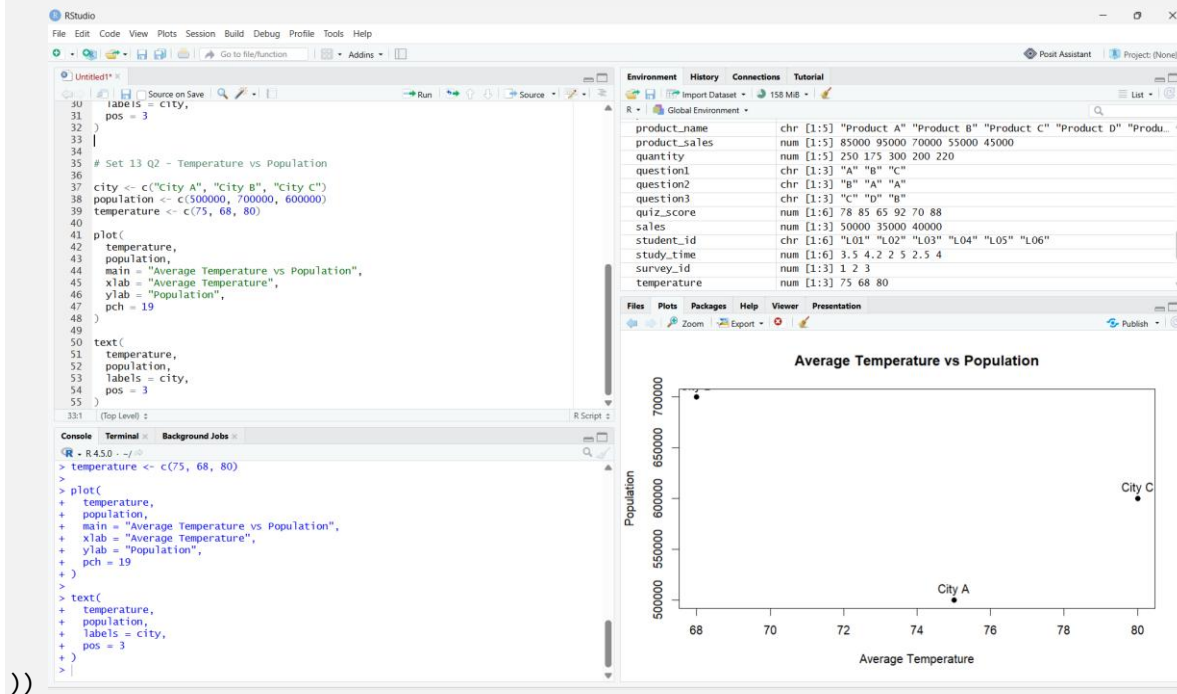
```
# # Set 13 Q2 - Temperature vs Population

city <- c("City A", "City B", "City C")
population <- c(500000, 700000, 600000)
temperature <- c(75, 68, 80)

plot(
  temperature,
  population,
  main = "Average Temperature vs Population",
  xlab = "Average Temperature",
  ylab = "Population",
  pch = 19
)

text(
  temperature,
  population,
  labels = city,
  pos = 3

```



Observation:

- ❑ City B has the **highest population (700000)** but a lower temperature of **68**.
- ❑ City C has the **highest temperature (80)** and population of **600000**.
- ❑ City A has a temperature of **75** and population of **500000**.
- ❑ With only three cities, there is **no clear strong relationship** between temperature and population.

PROGRAM 3: Geographic Data Table

To display the geographic information of each city in tabular form

```
# # Set 13 Q3 - Geographic Data Table
city <- c("City A", "City B", "City C")
population <- c(500000, 700000, 600000)
temperature <- c(75, 68, 80)
elevation <- c(1000, 800, 1200)
geographic_table <- data.frame(
  City = city,
  Population = population,
  Average_Temperature = temperature,
  Elevation = elevation
)
geographic_table
```

The screenshot shows the RStudio interface. The source editor contains R code for creating a data frame and plotting it. The console shows the execution of the code, resulting in a printed data frame.

```

46 ylab = "Population",
47 pch = 19
48 )
49
50 text(
51   temperature,
52   population,
53   labels = city,
54   pos = 3
55 )
56
57 # Set 13 Q3 - Geographic Data Table
58
59 city <- c("City A", "City B", "City C")
60 population <- c(500000, 700000, 600000)
61 temperature <- c(75, 68, 80)
62 elevation <- c(1000, 800, 1200)
63
64 geographic_table <- data.frame(
65   City = city,
66   Population = population,
67   Average_Temperature = temperature,
68   Elevation = elevation
69 )
70
71 geographic_table

```

Console output:

```

> city <- c("City A", "City B", "City C")
> population <- c(500000, 700000, 600000)
> temperature <- c(75, 68, 80)
> elevation <- c(1000, 800, 1200)
>
> geographic_table <- data.frame(
+   City = city,
+   Population = population,
+   Average_Temperature = temperature,
+   Elevation = elevation
+ )
>
> geographic_table
  City Population Average_Temperature Elevation
1 City A      5e+05                75      1000
2 City B      7e+05                68        800
3 City C      6e+05                80      1200
>

```

Observation

The table displays:

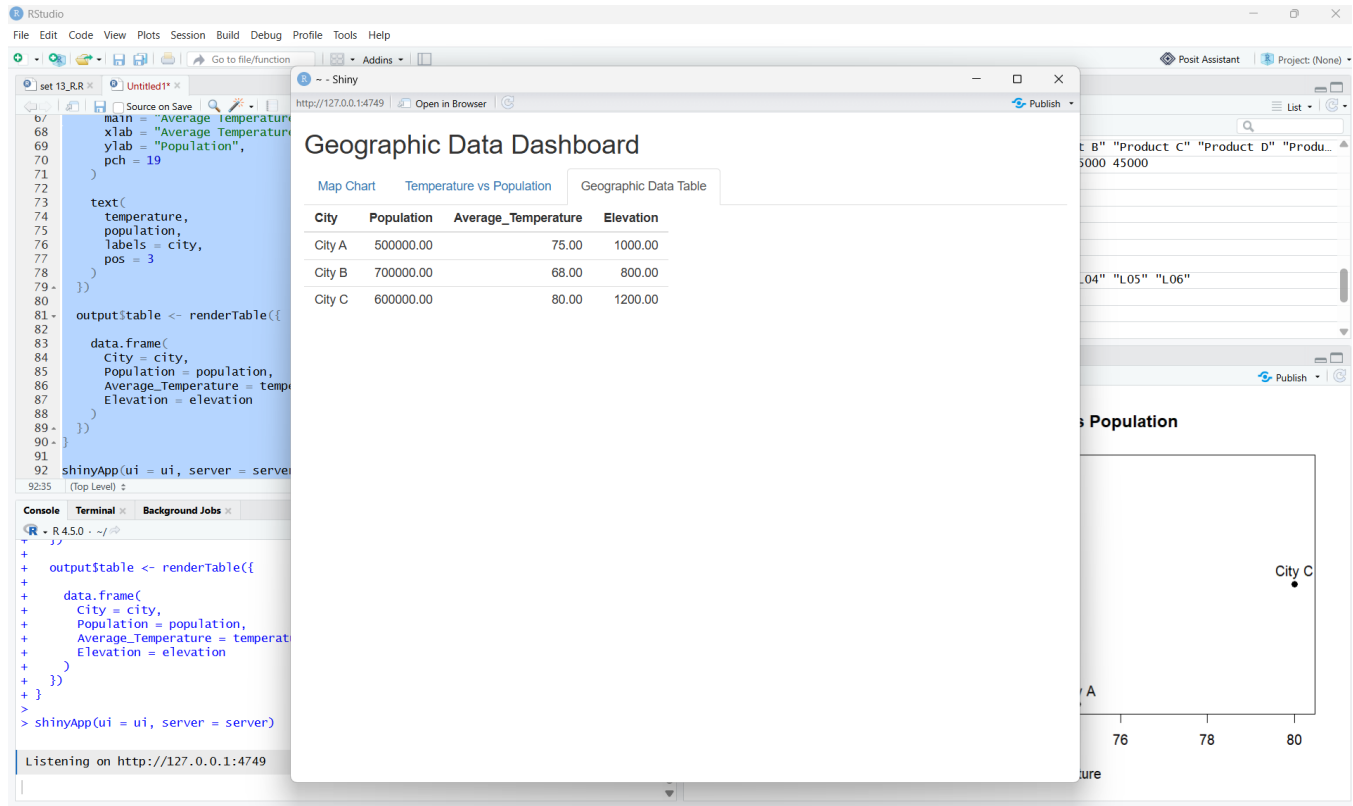
- **City**
- **Population**
- **Average Temperature**
- **Elevation**

PROGRAM 4: Interactive Dashboard in RStudio

To develop an interactive dashboard combining the geographic map-style chart, scatter plot and geographic data table.

```
plot(
  temperature,
  population,
  main = "Average Temperature vs Population",
  xlab = "Average Temperature",
  ylab = "Population",
  pch = 19
)
text(
  temperature,
  population,
  labels = city,
  pos = 3
)
})
output$table <- renderTable({

  data.frame(
    City = city,
    Population = population,
    Average_Temperature = temperature,
    Elevation = elevation
  )
})
}
shinyApp(ui = ui, server = server))
```



Output 4: Interactive Monthly Sales Dashboard

Observation: An interactive geographic dashboard was successfully developed in RStudio using Shiny.

Procedure:

- Create vectors for product categories and sales.
- Sort the sales values and use a horizontal bar chart as a funnel-style visualization.
- Use `data.frame()` to create the sales table.
- Use `pie()` to display the distribution of sales.
- Use Shiny to combine the funnel-style chart, pie chart, and table.
- Save each question as a separate R script.

Result:

The product category sales data was successfully analyzed using a funnel-style chart, pie chart, and table. An interactive dashboard was also developed in RStudio to explore the product category sales